

Form Number : _____

Paper Code
(0000CMA302119001)

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CLASSROOM CONTACT PROGRAMME
(Academic Session : 2019- 2020)

**COMPUTER
BASED TEST**
आसान है। 0

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

Test Type : SRG [COMMON TEST]

Test Pattern : AIIMS

TEST DATE : 15 - 09 - 2019

Important Instructions

Do not open this Test Booklet until you are asked to do so

1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only.

If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.

2. Duration of Test is 3 Hours and Questions Paper Contains 200 Questions. The Max. Marks are 200.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 1 marks, while one third mark will be deducted from the total of individual subject for each incorrect answer.

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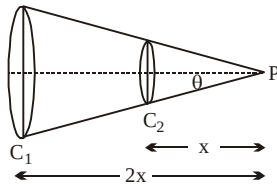
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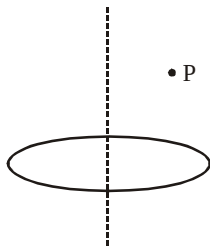
BEWARE OF NEGATIVE MARKING

1. The coils C_1 and C_2 have same number of turns and carry equal currents in the same sense. They subtend the same angle θ to P. If the magnetic field produced by C_1 at P is B then that produced by C_2 will be



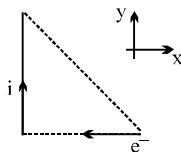
- (1) B (2) 2B (3) B/2 (4) None

2. A current carrying loop is lying on a horizontal plane as shown in figure current is anti-clockwise as seen from above the loop then direction of magnetic field at point P will be



- (1) ↙ (2) ↓ (3) ↗ (4) ↘

3. The direction of magnetic force on the electron as shown in the diagram is along

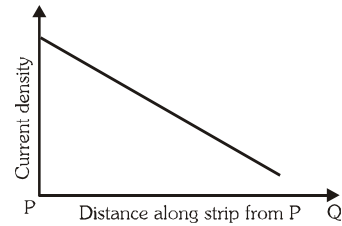


- (1) y-axis (2) -y-axis
(3) z-axis (4) -z-axis

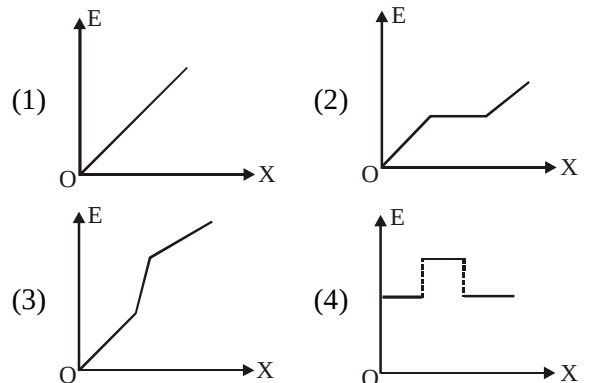
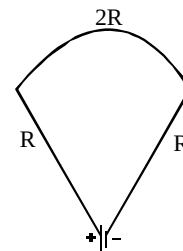
4. Carbon resistor has resistance specified by three bands having colour red, yellow and black. If this resistor is cut into two pieces of equal length then the new colour code of each one will be (Neglect tolerance of 4th band)

- (1) Brown, Red, Black
(2) Red, Orange, Black
(3) White, Blue, Black
(4) Black, Blue, Black

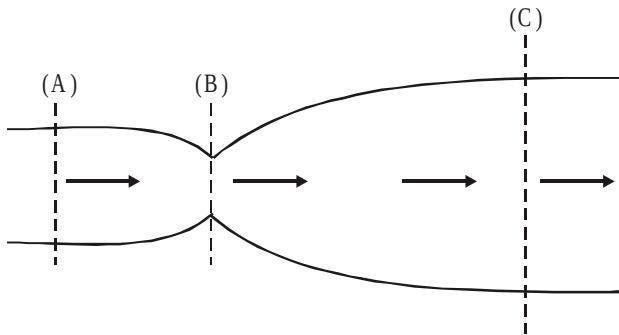
5. An electric current flows along an insulated strip PQ of a metallic conductor. The current density in the strip varies as shown in graph of figure. Which one of the following statements could explain this variation ?



- (1) The strip is narrower at P than at Q
(2) The strip is narrower at Q than at P
(3) The potential gradient along the strip is uniform
(4) The resistance per unit length of the strip is uniform
6. In which material do the conduction electrons have the largest mean time between collisions?
(1) Copper (2) Aluminium
(3) Nichrome (4) Tungsten
7. Three wires of equal length and of same material are connected to a battery as shown in figure. Which one of the following graphs represents the variation of electric field E in the conductor with distance x measured along the wire starting from positive terminal of the battery ?



8. Figure shows a copper wire of nonuniform cross section carrying current toward right for sectors A, B and C, select correct option.



- (1) $i_B > i_A > i_C$ (i = current)
 (2) $E_A > E_B > E_C$ (E = electric field)
 (3) $V_C > V_A > V_B$ (V = drift velocity)
 (4) $J_B > J_A > J_C$ (J = current density)
9. The wire shown in figure has a uniform cross-section A.

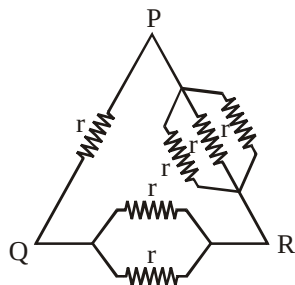
$$\overline{\quad\quad\quad} \quad x=0 \quad\quad\quad x=L$$

Resistivity of the material of wire is given by

$$\rho = \rho_0 \left(\frac{L}{L+x} \right)$$

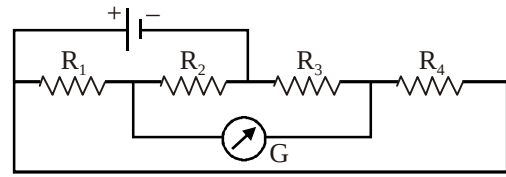
A potential difference V is applied across the wire :-

- (1) Resistance of wire is $\frac{\rho_0 L}{A} \cdot \ln(2)$
 (2) Current density is variable inside the wire
 (3) Electric field at $x = 0$ is $\frac{2V}{(\ln 2)L}$
 (4) Electric field at $x = L$ is $\frac{V}{(\ln 2)L}$
10. Six equal resistance are connected between points P, Q and R as shown in figure. Then net resistance will be minimum between :

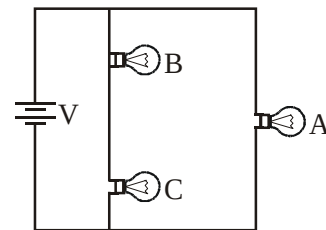


- (1) P and R
 (2) P and Q
 (3) Q and R
 (4) Any two points

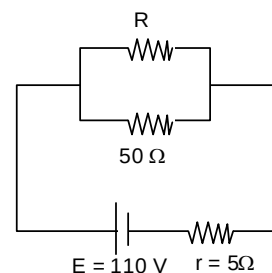
11. In the given circuit, the galvanometer G will show zero deflection if



- (1) $R_1 R_2 = R_3 R_4$
 (2) $R_1 R_3 = R_2 R_4$
 (3) $R_1 R_4 = R_2 R_3$
 (4) none of the above
12. Wattage of three bulbs A, B and C rated at same voltage are W_A , W_B and W_C respectively. Their brightness are represented by P_A , P_B and P_C . If $P_A = P_B = P_C$, then correct ratio of their wattage ($W_A : W_B : W_C$) is :-

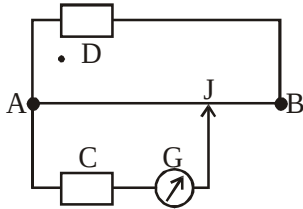


- (1) 1 : 2 : 3
 (2) 1 : 2 : 2
 (3) 1 : 4 : 4
 (4) 2 : 2 : 1
13. An electric circuit consists of a battery of emf $E = 110$ V, and internal resistance is 5Ω and two resistors connected in parallel to the source as shown in figure. Determine the value of R so that maximum power get dissipated in resistance R .

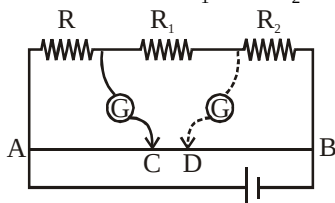


- (1) 50/11
 (2) 50/12
 (3) 50
 (4) 50/13

14. The figure shows a potentiometer arrangement. D is the driving cell, C is the cell whose e.m.f. is to be determined. AB is the potentiometer wire and G is a galvanometer. J is a sliding contact which can touch any point on AB. Which of the following are essential conditions for obtaining balance ?



- (A) The e.m.f. of D must be greater than the e.m.f. of C
(B) The positive terminals of D and C both must be joined to A
(C) The resistance of G must be less than the resistance of AB
(D) Resistance of wire AB should be very small
(1) Only A (2) Only A and C
(3) Only C and D (4) Only A, B and C
15. In a meter bridge experiment null point is obtained at 20 cm from one end of the wire when resistance x is balanced against another resistance y. If $x < y$, then where will be the new position of the null point from the same end. If one decides to balance a resistance of $4x$ against y :-
(1) 50 cm (2) 80 cm
(3) 40 cm (4) 70 cm
16. The diagram shows a modified meter bridge, which is used for measuring two unknown resistances at the same time. When only the first galvanometer is used, for obtaining the balance point, it is found at point C. Now the first galvanometer is removed and the second galvanometer is used, which gives balance point at D. Using the details given in the diagram, find out the value of R_1 and R_2 :-



$AB = L$; $AC = L/4$ and $AD = 2L/3$

- (1) $R_1 = 5R/3$, $R_2 = 4R/3$ (2) $R_1 = 4R/3$, $R_2 = R$
(3) $R_1 = 4R/3$, $R_2 = 5R/3$ (4) $R_1 = R$, $R_2 = R/3$

17. A constant potential difference is applied to the ends of a graphite rod, whose resistance decreases with a rise of temperature. The rod can be (1) covered with asbestos having low thermal conductivity or (2) left open to atmosphere. Answer for steady state.
(1) in both cases power lost is same but temperature of rod is more in case-1
(2) in case-1 power lost as well as temperature of rod is more
(3) in case-2 power lost as well as temperature of rod is more
(4) in case-1 power lost is more but temperature of rod is less

Paragraph for Question 18 to 20

When a current i flows through a resistance R for time t , it produces heat, $H = i^2 R t$ known as Joule's heat.

$$\text{Power consumed } P = \frac{H}{t} = i^2 R = \frac{V^2}{R} = Vi$$

So, if resistances are connected in series (i.e., i remains same) $P \propto R$ with $V \propto R$

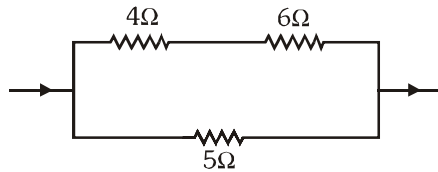
which implies that potential difference and power consumed will be more in larger resistance.

However, if resistances are connected in parallel (i.e., V is same) $P \propto \frac{1}{R}$ with $i \propto \frac{1}{R}$

which means that in parallel circuit, current and power consumed will be more in smaller resistance.

18. The water in an electric kettle begins to boil in 15 minutes after being switched on. If the water is to boil in 10 minutes using the main supply, the length of the heating element be
(1) Increased to $\frac{3}{2}$ of its initial value
(2) Decreased to $\frac{2}{3}$ of its initial value
(3) Increased to $\frac{5}{4}$ of its initial value
(4) Decreased to $\frac{4}{5}$ of its initial value
19. n identical bulbs, each designed to draw power P from a certain voltage supply, are joined in series across that supply. The total power which they will draw is
(1) nP (2) P (3) P/n (4) P/n^2

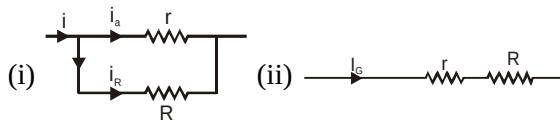
20. In the circuit shown, the heat produced in 5Ω resistor due to current flowing in it is 10 cal/sec. The heat generated in 4Ω resistor is



- (1) 1 cal/sec (2) 2 cal/sec
(3) 3 cal/sec (4) 4 cal/sec

Paragraph for Question 21 to 23

Measurements and interpretations of voltage and electric current signals are common in modern medicine. Occasionally, a situation arises in which a voltmeter or an ammeter is needed but it is not available. A galvanometer is an instrument that can be used to construct an ammeter (for measuring electric currents). It can also be used to construct a voltmeter (to measure voltages). In both cases, a resistor R must be connected to the galvanometer to effect the change.



To turn the galvanometer into an ammeter, the resistor R is connected in parallel, as shown in figure (i). The resistor R is connected in series with the galvanometer in order to turn it into a voltmeter, as shown in figure (ii). The current required to produce a full scale deflection in a galvanometer is 10 mA. The internal resistance of the galvanometer is 100Ω . Let V_r be the voltage across r (internal resistance of galvanometer) and V_R the voltage across R .

21. Which of the following formulas correctly applies to the ammeter circuit?
- (1) $V_r > V_R$
(2) $V_r < V_R$
(3) $V_r = V_R$
(4) more information is needed
22. What resistance must be connected in parallel to the galvanometer to turn it into an ammeter capable of reading electric currents up to 10.01 A?
- (1) 0.1Ω (2) 1Ω
(3) 10Ω (4) None of these

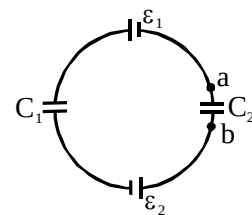
23. What resistance R must be connected in series to the galvanometer in order to convert it to a 100 V voltmeter?

- (1) 900Ω (2) 1000Ω
(3) 9900Ω (4) 10000Ω

24. In a parallel square plate air capacitor, a cathode beam comprising $n = 10^6$ electrons is emitted with a velocity $v_0 = 10^8$ m/s into the space between the plates. The potential difference between the plate is $\phi = 400$ V, the separation between the plates is $d = 2$ cm and the area of each plate is $\ell^2 = 100$ cm². The deflection of the electron beam, is :-

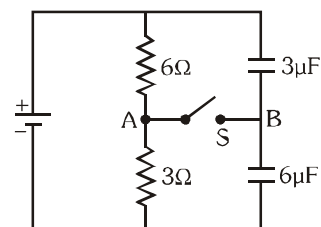
- (1) 1.6 mm (2) 1.76 mm (3) 0 mm (4) 5 mm

25. Find the potential difference across point a and b in fig.:-



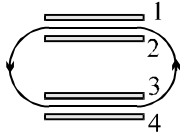
- (1) $\left(\frac{\epsilon_1 + \epsilon_2}{C_1 + C_2}\right) C_1$ (2) $\left(\frac{\epsilon_1 + \epsilon_2}{C_1 + C_2}\right) C_2$
(3) $\left(\frac{\epsilon_1 - \epsilon_2}{C_1 + C_2}\right) C_1$ (4) $\left(\frac{\epsilon_1 - \epsilon_2}{C_1 + C_2}\right) C_2$

26. In the circuit shown, when the switch S is closed, then

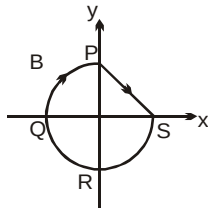


- (1) no charge flows through S
(2) charge flows from A to B
(3) charge flows from B to A
(4) charge flows initially from A to B and later from B to A

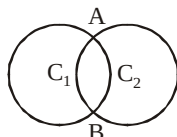
27. Figure shows the path of an electron in a region of uniform magnetic field. The path consists of two straight sections, each between a pair of uniformly charged plates, and two half-circles. The plates are named 1, 2, 3 & 4. Then



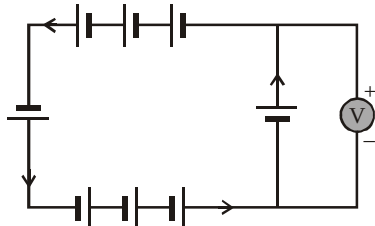
- (1) 1 and 3 at higher (positive) potential and 2 and 4 at lower (negative) potential
(2) 1 and 3 at lower potential and 2 and 4 at higher potential
(3) 1 and 4 at higher potential and 2 and 3 at lower potential
(4) 1 and 4 at lower potential and 2 and 3 at higher potential
28. A current carrying loop is placed in a uniform magnetic field pointing in negative z direction. Branch PQRS is a three quarter circle, while branch PS is straight. If force on branch PS is F , force on branch PQR is



- (1) $\sqrt{2}F$ (2) $\frac{F}{\sqrt{2}}$ (3) $\frac{\pi F}{\sqrt{2}}$ (4) $\sqrt{2}\pi F$
29. Two circular rings of identical radii & resistance = $36\ \Omega$ each are placed in such a way that they cross each other at centres C_1 & C_2 . A battery of 20 V is connected across A & B. Power delivered by battery is

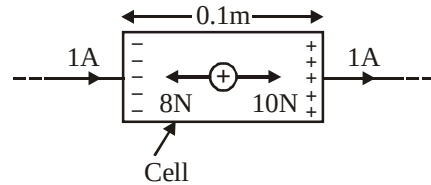


- (1) 50 W (2) 100 W (3) 200 W (4) 400 W
30. In the circuit shown in figure each battery is 5V and has an internal resistance of $0.2\ \Omega$. What is the reading of ideal voltmeter V (in Volt)?

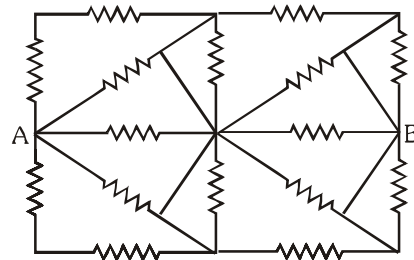


- (1) 0 (2) 1 (3) 2 (4) 3

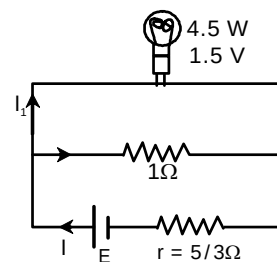
31. Figure shows a cell in which unit positive charge experiences a constant non electric force of 10N and a constant electric force of 8N in directions shown in the figure. The internal resistance of the cell is $r\ \Omega$. Then value of r will be



- (1) $2\ \Omega$ (2) $0.2\ \Omega$
(3) $8\ \Omega$ (4) $0.8\ \Omega$
32. The deflection of a moving coil galvanometer falls from 60 divisions to 12 divisions for the same value of current in the circuit, when a shunt of $12\ \Omega$ is connected. The resistance of the galvanometer is :-
- (1) $12\ \Omega$ (2) $24\ \Omega$
(3) $36\ \Omega$ (4) $48\ \Omega$
33. In the given circuit all resistors are identical. Their resistance is equal to $20\ \Omega$. Find the equivalent resistance between A & B.



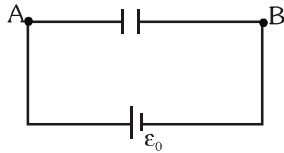
- (1) $9\ \Omega$ (2) $8\ \Omega$
(3) $20\ \Omega$ (4) $10\ \Omega$
34. A torch bulb rated at 4.5 W, 1.5 V is connected as shown in the figure. To make the bulb glow at full intensity, what should be the emf of the cell?



- (1) 4.5 V (2) 1.5 V
(3) 3 V (4) 9 V

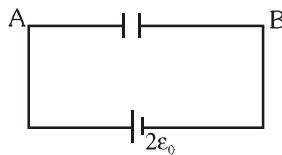
35. In Allen Institute one student wants to perform an experiment with two battery and one capacitor

Event Ist



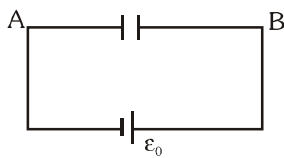
After event Ist

Event IInd



After event IInd

Event III

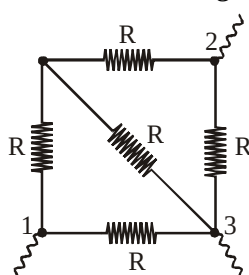


Work done by the battery in event IInd is α &

in event IIIrd is β then find the value of $\left| \frac{2\beta}{\alpha} \right|$?

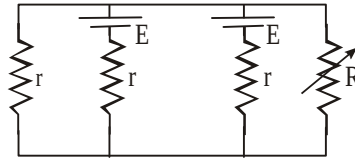
- (1) 3 (2) 2 (3) 1 (4) 4

36. The resistances in a burner of an electric stove are connected together in an arrangement shown in figure. This arrangement is connected to the supply mains at points 1 and 2, and after a certain time, 500 grams of water are heated to the boiling point. How much water can be heated to the boiling point in the same time interval when the arrangement of resistances is connected to the supply mains at points 1 and 3? The initial temperature of the water is the same in both cases. Neglect all heat losses.



- (1) 500 g (2) 200 g (3) 800 g (4) 333 g

37. In shown circuit there are two identical ideal cells of emf E and $r = 9 \Omega$, find the value of R (in Ω) so that the power dissipated in resistance R is maximum.



- (1) 9Ω (2) 6Ω
(3) 13.5Ω (4) 3Ω

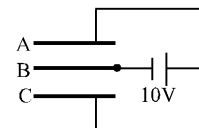
38. 2 conducting objects one with charge of $+Q$ and another with $-Q$ are kept on x-axis at $x = 0$ and $x = 1$ respectively. The electric field on the

x-axis is given by $3Q \left(x^2 + \frac{4}{3} \right)$. The

capacitance of this configuration of objects is.

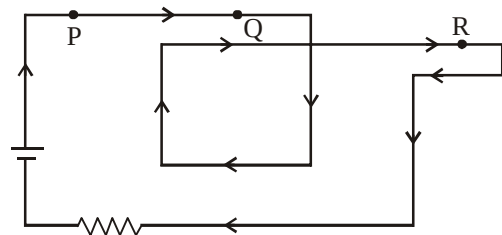
- (1) 0.1 F (2) 0.2 F
(3) 0.4 F (4) 0.5 F

39. Three plates A, B and C each of area 0.1 m^2 are separated by 0.885 mm from each other as shown in the figure. A 10 V battery is used to charge the system. The energy stored in the system is



- (1) $1 \mu\text{J}$ (2) $10^{-1} \mu\text{J}$
(3) $10^{-2} \mu\text{J}$ (4) $10^{-3} \mu\text{J}$

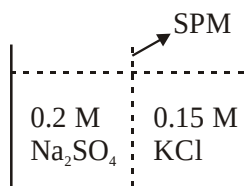
40. A battery establishes a steady current around the circuit shown. A compass needle is placed successively at points P, Q, and R, just above the wire (slightly out of the plane of the page). The relative deflection of the needle, in descending order, is



- (1) P, Q, R. (2) Q, R, P.
(3) R, Q, P. (4) Q, P, R.

41. The molarity of 0.3 N $\text{Ca}_3(\text{PO}_4)_2$ solution will be:-
(1) 1.8 M (2) 0.3 M (3) 0.05 M (4) 5 M
42. Which will form maximum boiling azeotrope?
(1) $\text{C}_6\text{H}_6 + \text{C}_6\text{H}_5\text{CH}_3$
(2) $\text{HNO}_3 + \text{H}_2\text{O}$
(3) $\text{C}_2\text{H}_5\text{OH} + \text{H}_2\text{O}$
(4) n-hexane + n-heptane
43. van't Hoff factor of Hg_2Cl_2 in its solution will be. If Hg_2Cl_2 is 70% ionized in the solution :-
(1) 1.6 (2) 2.4 (3) 3 (4) 3.6
44. Correct statement is/are :-
(1) RBC placed in pure water will swell because of endo-osmosis.
(2) Ethylene glycol is used as anti freeze for radiator of car.
(3) K_b and K_f constants depend on nature of solvent
(4) All are correct.
45. A complex is written as $\text{M}(\text{en})_y \cdot x\text{Br}$. Its 0.05 molar solution shows 2.46 atm osmotic pressure at 27°C . Assuming 100% ionised and coordination number of metal (III) is six, complex may be?
(1) $[\text{M}(\text{en})_2\text{Br}_2]\text{Br}$ (2) $[\text{M}(\text{en})_3]\text{Br}_3$
(3) $[\text{M}(\text{en})_2\text{Br}]\text{Br}_2$ (4) $[\text{M}(\text{en})]\text{Br}_3$

46.

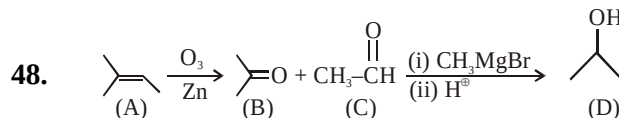


Choose the correct options ?

- (A) Osmotic pressure of Na_2SO_4 solution is lesser than osmotic pressure of KCl solution
(B) Water will flow from KCl to Na_2SO_4 solution
(C) Water will flow from Na_2SO_4 solution to KCl solution
(D) Osmotic pressure of Na_2SO_4 solution is higher than the osmotic pressure of KCl solution
- (1) A, B (2) B, D
(3) C, D (4) None of these

47. (A) $\xrightarrow[\text{CHCl}_3]{\text{KOH}}$ (B) $\xrightarrow[\text{H}^+]{\text{LiAlH}_4}$ $\text{CH}_3\text{CH}_2\text{-NH-CH}_3$
Identify compound 'A' :-

- (1) $\text{CH}_3\text{-CH}_2\text{-C(=O)-OH}$ (2) $\text{CH}_3\text{-CH}_2\text{-C(=O)-NH}_2$
(3) $\text{CH}_3\text{-CH}_2\text{-C(=O)-Cl}$ (4) $\text{CH}_3\text{-CH}_2\text{-NH}_2$

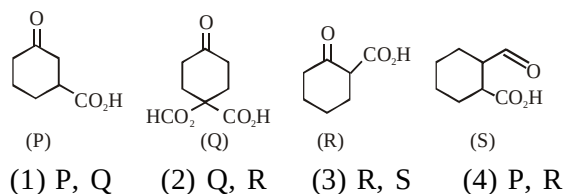


Compound (B) and (C) is differentiated by :-

- (1) Iodoform test (2) 2,4 DNP
(3) Tollen's reagent (4) All
49. Which of the following gives positive iodoform test ?

- (1) Ph-CHO (2) Ph-C(=O)-Ph
(3) Ph-C(=O)-CH_3 (4) $\text{Ph-C(=O)-CH}_2\text{-CH}_3$

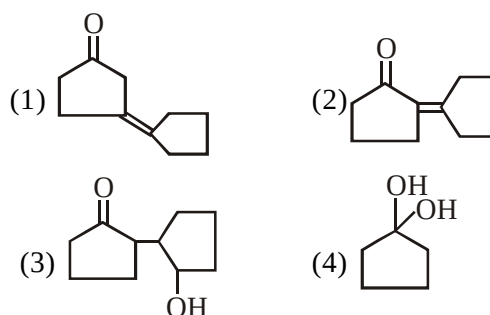
50. Which of the following easily undergoes decarboxylation on heating ?



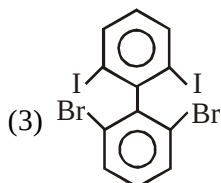
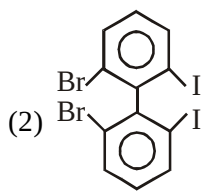
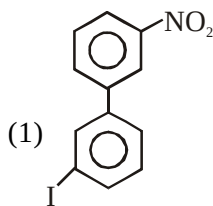
51. $\text{Ph-C(=O)-NH}_2 \xrightarrow[\text{P}_2\text{O}_5/\Delta]{\text{POCl}_3 \text{ or}}$ (A) Product (A) is :-

- (1) Ph-NH_2 (2) $\text{Ph-CH}_2\text{-NH}_2$
(3) Ph-CH(OH)-NH_2 (4) $\text{Ph-C}\equiv\text{N}$

52. $\xrightarrow[\Delta]{\text{dil KOH}}$ Product (A) is :-

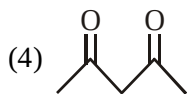
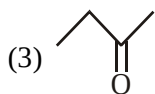
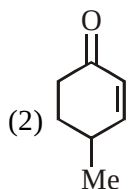
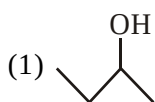


53. Which of the following biphenyl is optical active :-

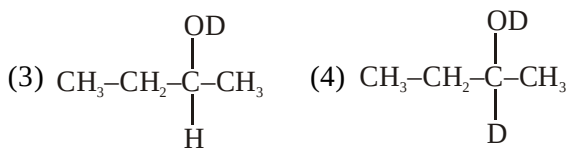
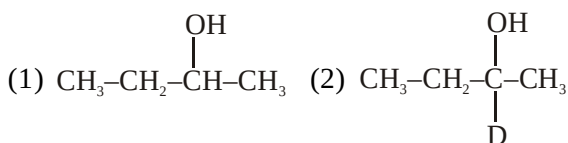
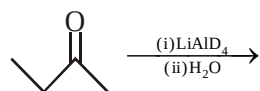


(4) None

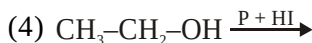
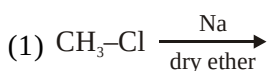
54. Which of the following can show both tautomerism and optical isomerism :-



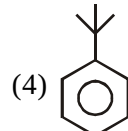
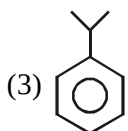
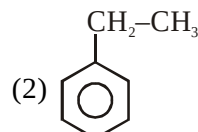
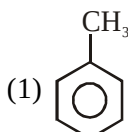
55. Give the product of the following reaction :-



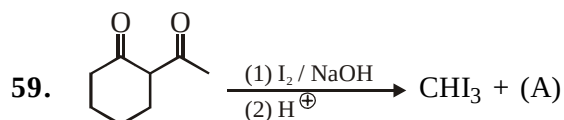
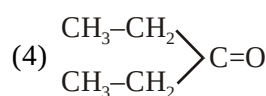
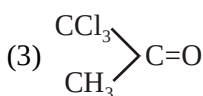
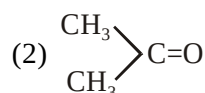
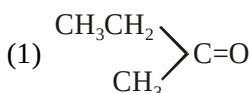
56. Which of the following reaction does not give ethane as product ?



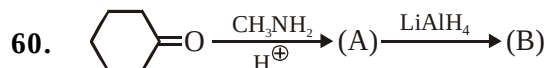
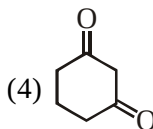
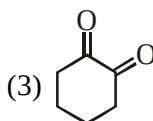
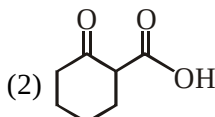
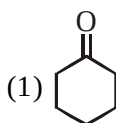
57. Which of the following doesn't give benzoic acid on oxidation using hot acidic KMnO₄ ?



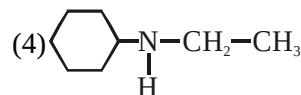
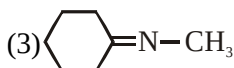
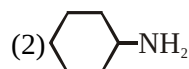
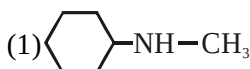
58. Most reactive species towards NAR is :-



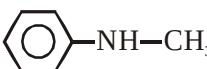
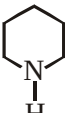
What is A?



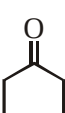
What will be (B)?



61. Which will give carbylamine reaction?

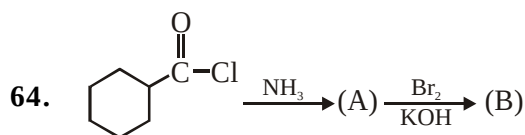
- (1) $(\text{CH}_3)_2\text{CH}-\text{NH}_2$
 (2) $(\text{CH}_3)_2\text{CH}-\text{NH}-\text{CH}_3$
 (3) 
 (4) 

62. An organic compound (A) gives iodoform test and gives 2, 4-DNP test. Compound (A) will be :

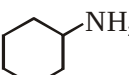
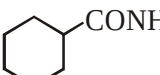
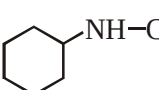
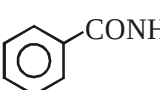
- (1) $\text{CH}_3\text{CH}_2\text{C}(=\text{O})\text{H}$ (2) $\text{CH}_3\text{C}(=\text{O})\text{H}$
 (3)  (4) $\text{CH}_3\text{C}(=\text{O})\text{Cl}$

63. Which is correct order of +I effect

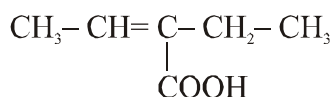
- (1) $-\text{CH}(\text{CH}_3)_2 > -\text{C}(\text{CH}_3)_3 > -\text{CH}_2\text{CH}_3$
 (2) $-\text{COO}^- > -\text{CH}(\text{CH}_3)_2 > -\text{CT}_3$
 (3) $-\text{CH}_3 > -\text{CD}_3 > -\text{CT}_3$
 (4) $-\text{O}^- > -\text{CH}_3 > -\text{COO}^-$



What will be "B"?

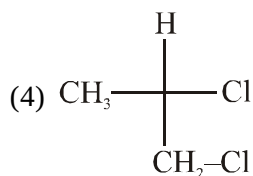
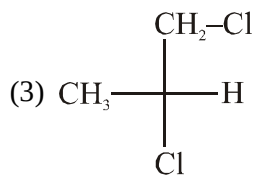
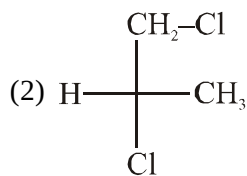
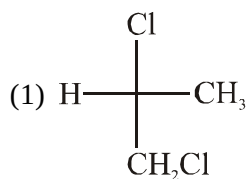
- (1)  (2) 
 (3)  (4) 

65. The IUPAC name of given compound is :-

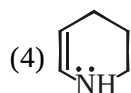
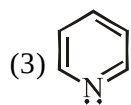
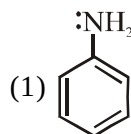


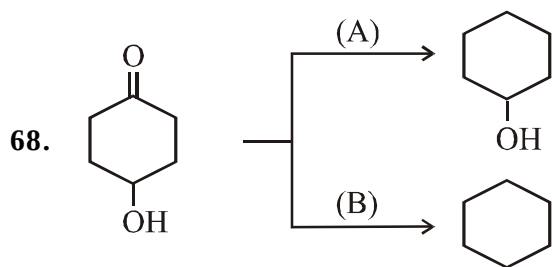
- (1) 3-Carboxy-2-pentene
 (2) 3-Ethyl-2-buten-4-oic acid
 (3) 2-Ethyl-2-butenic acid
 (4) 2-Ethylidene butanoic acid

66. Which of the following compound has S-configuration :-



67. Find out structure in which lone pair is Localised.





Suitable reagent (A) and (B) will be respectively :

- (1) LiAlH_4 and Red P + HI
 (2) Red P + HI and Red P + HI
 (3) LiAlH_4 and Zn-Hg/HCl
 (4) $\text{NH}_2\text{-NH}_2/\text{OH}^\ominus$ and Red P + HI
69. $\text{CH}_3\text{-NH}_2 + \text{CHCl}_3 + \text{KOH} \longrightarrow (\text{x}) + \text{KCl} + \text{H}_2\text{O}$
 x is :-

- (1) $\text{CH}_3\text{-CN}$
 (2) $\text{CH}_3\text{-NH-CH}_3$
 (3) $\text{CH}_3\text{-NC}$
 (4) $\text{CH}_3\text{-CH}_2\text{-NC}$

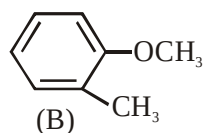
70. IUPAC Names of some compounds are given.
 Which one is incorrect -

- (1)
 3-Methyl-4-ethyl-heptane
 (2)
 3-Methyl-2-butanol
 (3)
 2-Ethyl-3-methyl-but-1-ene
 (4) $\text{CH}_3\text{-C}\equiv\text{C-CH(CH}_3)_2$
 4-Methyl-2-pentyne

71. Which of the following can show geometrical isomerism :-

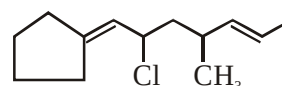
- (1)
 (2)
 (3)
 (4)

72. Metamer of compound B is :-



- (1)
 (2)
 (3)
 (4)

73. The number of stereoisomers possible for the following compound

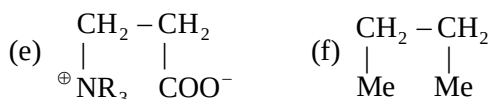
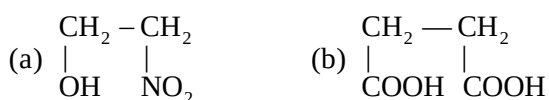


- (1) 2
 (2) 4
 (3) 8
 (4) 16

74. Minimum number of carbon required to show chain isomerism and position isomerism in aldehyde respectively :-

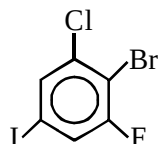
- (1) 4, 5
(2) 4, 4
(3) 5, 5
(4) 5, 4

75. How many following compound have Gauche Conformation is more stable than anti (staggered) conformation :

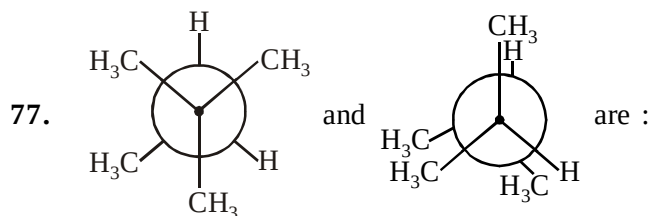


- (1) 6 (2) 5
(3) 4 (4) 3

76. IUPAC name for the given compound is :

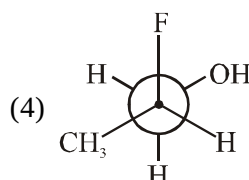
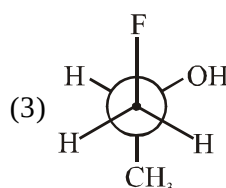
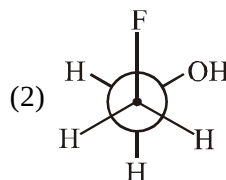
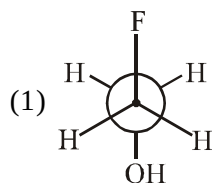


- (1) 1-Bromo-2-chloro-4-iodo-6-fluorobenzene
(2) 2-Bromo-1-chloro-3-fluoro-5-iodobenzene
(3) 2-Bromo-3-chloro-1-fluoro-5-iodobenzene
(4) 1-Bromo-6-chloro-2-fluoro-4-iodobenzene

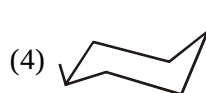
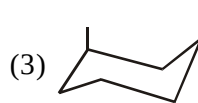
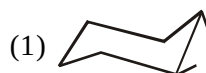


- (1) Conformational isomer
(2) Position isomer
(3) Chain isomer
(4) Identical molecule

78. Which conformer is most stable of 2-Fluoroethanol :-



79. Which is cis-1,2-dimethyl cyclohexane in following conformers :-



80. Find out the group which not show mesomeric effect but show I effect

- (1) $-\ddot{\text{O}}\text{H}$
(2) $-\text{C}\equiv\text{N}$
(3) $-\text{NH}_3^+$
(4) $-\text{C}(=\text{O})-\text{OH}$

81. Which cross will not produce homozygous and heterozygous progenies in equal proportion ?

- (1) $Tt \times tt$ (2) $Tt \times Tt$
(3) $TT \times Tt$ (4) $TT \times tt$

82. A cross between plants heterozygous for two different genes ($AaBb \times AaBb$), each assorting independently. What fraction of progeny will show recessive phenotype for atleast one gene?

- (1) $1/16$ (2) $9/16$
(3) $7/16$ (4) $3/16$

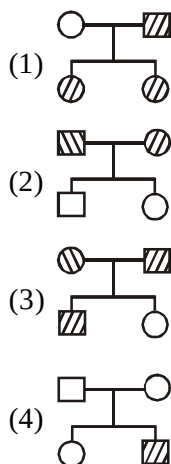
83. Source of non-parental phenotypes in mendelian dihybrid cross is :-

- (1) Mutation
(2) Dominance
(3) Crossing-over
(4) Independent Assortment

84. Phenotype of Recessive allele is seen in :-

- (1) Homozygous condition
(2) Heterozygous condition
(3) Hemizygous condition
(4) Both 1 and 3

85. Which pedigree is not possible for myotonic dystrophy ?



86. F_1 generation when resembles :-
A : either of the two parents
B : inbetween of the two parents
C : with both the parents

Choose correct match :-

	A	B	C
1	Dominance	Incomplete dominance	Co-dominance
2	Dominance	Co-dominance	Incomplete dominance
3	Incomplete dominance	Co-dominance	Dominance
4	Co-dominance	Dominance	Incomplete dominance

87. If there are 6 multiple alleles of a gene then how many types of homozygous conditions are possible in population ?

- (1) 15 (2) 6
(3) 10 (4) can't be predicted

88. In an example of incomplete dominance, pure red flowered plant are crossed with pure white flowered plant and the F_1 individuals have pink flowers. Which one of the following is not correct ?

- (1) Pink flowered plant will produce only pink flowered offsprings
(2) Pink flowered plant will produce offsprings having three kinds of flowers, if self pollinated
(3) The genes of the hybrid pink flowered plant will segregate, if self pollinated
(4) Half of the offsprings of the pink flowered plant will be homozygous, if it was self-pollinated

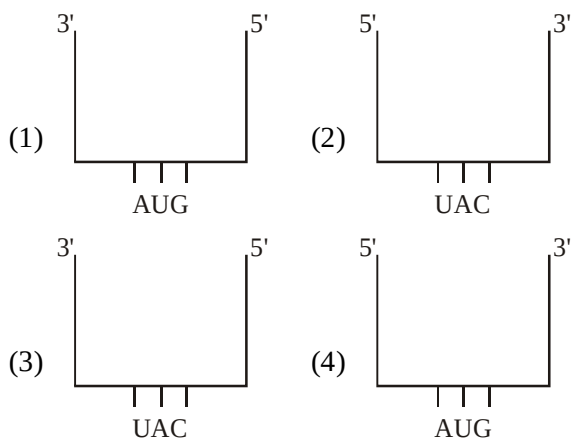
89. In a plant species, flower colour yellow is dominant over white and fruit shape round is dominant over elongated. Crossing was performed between two pure lines one having yellow flower and round fruit and another with white flower and elongated fruit. About 40 plants survived in F_1 progeny. Plants of F_1 were allowed to self fertilize and about 1920 plants survived in F_2 . If the traits follow Mendelian inheritance the number of plants would have yellow flower and round fruit in F_1 and F_2 are respectively :-

- (1) 40, 960 (2) 40, 540
(3) 40, 1920 (4) 40, 1080

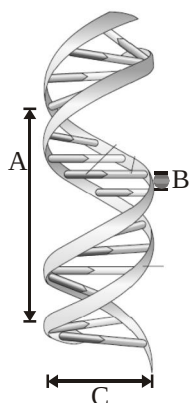
90. A plant can produce red flowers only when both gene P and Q are present in its genotype. Presence of only P or only Q result in blue flower. Complete absence of dominant genes result in white flower. A plant having genotype PpQq was self pollinated. What percentage of progeny will produce red flowers. :-

- (1) 56.25% (2) 18%
(3) 23.25% (4) 37.50%

91. Which one of the following would be the first t-RNA for protein synthesis :-



92. Given figure represents the DNA double helix model as proposed by Watson and Crick (1953). Select the option that shows **correct** measurements of A, B and C.



- (1) A-3.4 nm, B – 0.34 nm, C – 2 nm
(2) A-34 nm, B-3.4 nm, C – 20 nm
(3) A – 3.4 Å, B – 0.34 Å, C – 20 Å
(4) A – 34 Å, B – 3.4 Å, C – 2 Å

93. If the DNA of virus is labelled with ^{32}P and the protein coat of virus is labelled with ^{35}S after transduction which molecule would be present inside bacterial cell?

- (1) ^{32}P only
(2) ^{35}S only
(3) Both ^{35}S and ^{32}P
(4) Neither molecule would be present inside the cell.

94. Which of the following is Nucleoside ?

- (1) Adenylic Acid
(2) Adenosine, Uridine
(3) Adenine, Uracil
(4) Deoxy guanylic acid, Uridine

95. 5 Methyl uracil is always present in which Nucleic Acid

- (1) RNA and DNA
(2) Only DNA
(3) Only RNA
(4) Only double helical RNA

96. Which of the following is not true regarding the genome size of the given organisms :-

- (1) Bacteriophage $\phi \times 174$ – 5386 base pairs
(2) Bacteriophage lambda – 48502 base pairs
(3) E.coli – 4.6×10^6 base pairs
(4) Human (haploid content) – 3.3×10^9 base pairs

97. During replication of DNA, Okazaki fragments are synthesised in the direction of :-

- (1) $3' \rightarrow 5'$ (2) $5' \rightarrow 5'$
(3) $3' \rightarrow 3'$ (4) $5' \rightarrow 3'$

98. Which type of ribosomal RNA is not found in prokaryotes?

- (1) 23s (2) 16s (3) 5s (4) 5.8s

99. The discontinuously synthesised fragments of DNA are later joined by :-

- (1) DNA polymerase-I
(2) DNA polymerase-II
(3) DNA polymerase-III
(4) DNA ligase

- 100.** In DNA replication, the leading strand replicates in the :-
 (1) 5' → 3' direction discontinuously
 (2) 5' → 3 direction continuously
 (3) 3' → 5' direction discontinuously
 (4) 3' → 5' direction continuously
- 101.** Earth was present in which galaxy:-
 (1) Milky way (2) Cosmos
 (3) Aakash – Ganga (4) Both 1 and 3
- 102.** Which factors are not effect the Hardy–Weinberg equilibrium?
 (1) Gene migration and mutation
 (2) Genetic recombination and natural selection
 (3) Genetic drift, Gene flow
 (4) Large population, Random mating
- 103.** Which of the following is the correct sequence of events in the origin of life?
 I. Formation of protobionts
 II. Synthesis of organic monomers
 III. Synthesis of organic polymers
 IV. Formation of DNA-based genetic systems
 (1) I, III, II, IV
 (2) I, II, III, IV
 (3) II, III, IV, I
 (4) II, III, I, IV
- 104.** What is the main key concept of darwinian theory of evolution
 (A) Natural selection (B) Branching descent
 (C) Mutation (D) Genetic variation
 (1) A,C,D (2) A,B (3) A,B,C,D (4) A,D
- 105.** We are still not very clear that evolution is a process or the result of a process. But when we describe the story of this world we describe that evolution is a :-
 (1) Process
 (2) Consequence of natural selection
 (3) Unknown
 (4) Both 1 and 2
- 106.** Find out the correct sequence of evolution of Birds :-
 (1) Early reptiles – Sauropsids– therapsid– bird
 (2) Early reptiles – Synapsids–thecodont–birds
 (3) Early reptiles –Sauropsids–thecodont–birds
 (4) Early reptiles –Synapsids–sauropsids–birds
- 107.** According to Darwin, the founder of theory of natural selection fitness refers to :-
 (1) Inbuilt variation
 (2) Reproductive fitness
 (3) Struggle
 (4) Physical fitness
- 108.** Dinosaurs suddenly become extinct from earth about before how many years ?
 (1) 300 million years (2) 100 million years
 (3) 65 million years (4) 200 million years
- 109.** Before the industrial revolution in england in 1850, Which type of *Biston betularia* present in more amount :-
 (1) White winged (2) Dark winged
 (3) Red winged (4) All of the above
- 110.** Which one of the following are analogous structure?
 (1) Wings of Bat and Wings of Pigeon.
 (2) Gills of Prawn and Lungs of Man.
 (3) *Thorns of Bougainvillea* and Tendrils of *Cucurbita*
 (4) Flippers of Dolphin and Legs of Horse.
- 111.** Darwin judged the fitness of an individual by :-
 (1) Strategy to obtain food
 (2) Ability to defend itself
 (3) Number of offspring
 (4) Dominance over other individuals
- 112.** Darwin finches are not the example of :-
 (1) Genetic drift
 (2) Natural selection
 (3) Allopatric speciation
 (4) Adaptive convergence
- 113.** What is the main essence of Darwin theory of evolution :-
 (1) Genetic variation
 (2) Isolation
 (3) Struggle for existence
 (4) Natural selection

114. A process in which heritable variation enabling better survival are enabled to reproduce and leave greater number of progeny is called :-
 (1) Genetic drift (2) Natural selection
 (3) Founder effect (4) Both 1 and 3
115. Select the incorrect statements :-
 A. The work of malthus on population influenced Darwin
 B. Evolution is a directed process in the sense of determinism.
 C. The geological history of earth is not related with the biological history of earth.
 D. During evolution the rate of appearance of new life forms is linked to the life cycle .
 (1) A and B (2) B and C
 (3) A and D (4) B and D
116. In Galapagos island, Darwin found different varieties of finches, originated from a common ancestor. The original variety of these finches was:-
 (1) Insectivorous (2) Frugivorous
 (3) seed-eating (4) cactus eating
117. Appearance of teeth in the embryos of birds is an example of :-
 (1) Ontogeny repeats phylogeny
 (2) Speciation
 (3) Vestigial organs
 (4) Atavism
118. Excess use of herbicides, pesticides has only resulted in selection of resistance varieties in a much lesser time scale it was an example of evolution by :-
 (1) Mass extinction (2) Anthropogenic action
 (3) Continental drift (4) Both 1 and 3
119. Match the following column and choose the correct answer :-

Column I (Animal)	Column II (Time)
A. Jawless fishes	I. 320 mya
B. Sea weeds	II. 350 mya
C. Invertebrates	III. 200 mya
D. Ichthyosaurs	IV. 500 mya

 (1) A-II, B-I, C-IV, D-III
 (2) A-I, B-II, C-III, D-IV
 (3) A-II, B-IV, C-III, D-I
 (4) A-I, B-III, C-IV, D-II

120. Hugo de vries gave the theory of evolution was mainly based on :-
 (1) Survival of fittest
 (2) Natural selection
 (3) Mutation
 (4) Inheritance of acquired character

DIRECTIONS FOR Q. NO. 121 TO 180

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- A. If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 B. If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 C. If Assertion is True but the Reason is False.
 D. If both Assertion & Reason are False.

121. **Assertion** :- Electrolyte are more conducting than metal.
Reason :- Ions moves faster than electrons
 (1) A (2) B (3) C (4) D
122. **Assertion** :- A voltmeter is an inherently inaccurate instrument.
Reason :- A voltmeter is always connected in parallel in a circuit.
 (1) A (2) B (3) C (4) D
123. **Assertion** :- Thick metallic strips are used to join wires in a potentiometer.
Reason :- Metallic strips are used to provide mechanical support to potentiometer wires.
 (1) A (2) B (3) C (4) D
124. **Assertion** : Fuse wire should have low melting temperature and high resistivity.
Reason: Fuse saves an appliance from damage by increasing its current passing capacity.
 (1) A (2) B (3) C (4) D
125. **Assertion** : A larger dry cell has higher emf.
Reason : The emf of a dry cell is proportional to its size.
 (1) A (2) B (3) C (4) D

126. Assertion : A bird sitting on a high power line (say 11kV) is not affected practically.

Reason : Because no current flows through birds body, the circuit being incomplete.

- (1) A (2) B (3) C (4) D

127. Assertion :- If capacitor is filled with, same thickness (t) of dielectric ($t < d$) and conducting sheet one after another, then capacitance are C_1 and C_2 respectively then $C_1 < C_2$

Reason :- Capacitance is more in presence of metal sheet in compare to dielectric sheet as $K_{\text{Conduct}} > K_{\text{dielectric}}$

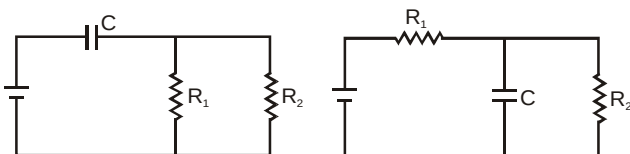
- (1) A (2) B (3) C (4) D

128. Assertion :- Circuits containing high capacity capacitors, charged to high voltage should be handled with caution, even when the current in the circuit is switched off.

Reason :- When an isolated capacitor is touched by hand or any other part of the human body, there is an easy path to the ground available for the discharge of the capacitor.

- (1) A (2) B (3) C (4) D

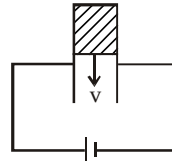
129. Assertion :- Time constants for charging of the circuits shown in the figure are different.



Reason :- Instantaneous current through the capacitor branch is same at any instant for both the circuits, if batteries are inserted in the circuits at $t = 0$

- (1) A (2) B (3) C (4) D

130.



Assertion :- If a dielectric is inserted between plates of capacitor with constant velocity a constant current is flowing in circuit.

Reason :- Rate of change of capacitance is constant.

- (1) A (2) B (3) C (4) D

131. Assertion :- Although water has very large dielectric constant $K = 80.4$ but it not commonly used as a dielectric in capacitors.

Reason :- Water can ionize certain materials, so some ions may enter into water increasing the conductivity of water, making water a bad dielectric.

- (1) A (2) B (3) C (4) D

132. Assertion :- A dielectric slab is slightly inserted in charged parallel plate capacitor and then released slab will execute oscillation.

Reason :- Electrostatic field is conservative field.

- (1) A (2) B (3) C (4) D

133. Assertion : A car battery is of 12 V. Eight dry cells of 1.5 V connected in series can give 12 V. Still such cells are not used in starting a car.

Reason : It is easier to start a car engine on a warm day than on a rainy day.

- (1) A (2) B (3) C (4) D

134. Assertion :- The angle of dip at geomagnetic pole is zero

Reason :- In a vertical plane perpendicular to the magnetic meridian the dip needle becomes exactly vertical

- (1) A (2) B (3) C (4) D

135. Assertion :- Magnetic force between two charge is generally much smaller than the electric force between them.

Reason :- Speeds of charges are much smaller than the free-space speed of light.

- (1) A (2) B (3) C (4) D

136. Assertion :- In current carrying wires, magnetic force considerably exceeds the electric forces acting on excess charges of the wires.

Reason :- The excess charges on the wires are negligibly small in comparison with the total-charge of carriers.

(1) A (2) B (3) C (4) D

137. Assertion :- Magnetic field arises due to charge in motion but a system may have magnetic moment even though its net charge is zero.

Reason :- Uncharged magnetic material may have dipole moment.

(1) A (2) B (3) C (4) D

138. Assertion :- To locate the magnetic meridian of a certain place, the vertical scale of a dip circle is first rotated till its needle becomes vertical then the vertical circle is exactly turned by 90° from this position

Reason :- At a certain place, the apparent angle of dip in vertical plane perpendicular to the magnetic meridian is 90°

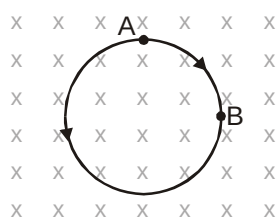
(1) A (2) B (3) C (4) D

139. Assertion :- The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic material as a core inside the coil.

Reason :- Soft iron has a high magnetic permeability and cannot be easily magnetized or demagnetized.

(1) A (2) B (3) C (4) D

140. Assertion :- In a region of time varying magnetic field, two charges of same magnitude moves from A to B in clockwise and anticlockwise manner. Work done in clockwise path is equal to work done in anticlockwise path.



Reason :- Work done in electric field is independent of path.

(1) A (2) B (3) C (4) D

141. Assertion :- 0.1 M aqueous solution of glucose has same depression in freezing point as 0.1 M aqueous solution of urea.

Reason :- K_f for both has different values.

(1) A (2) B (3) C (4) D

142. Assertion :- ΔV_{mix} and ΔS_{mix} for an Ideal solution is zero.

Reason :- A-B interaction in an ideal solution are same as between A-A and B-B.

(1) A (2) B (3) C (4) D

143. Assertion :- Relative lowering of vapour pressure is equal to mole fraction of the solute.

Reason :- Relative lowering in vapour pressure is a colligative property.

(1) A (2) B (3) C (4) D

144. Assertion :- One molar solution is always more concentrated than one molal solution.

Reason :- The amount of solvent in 1M and 1m aqueous solution is not equal.

(1) A (2) B (3) C (4) D

145. Assertion :- H-COOH and H-COOCH_3 are homologous.

Reason :- Homologous differ by atleast one " CH_2 " unit.

(1) A (2) B (3) C (4) D

146. Assertion :- Aniline on reaction with NaNO_2/HCl at 0°C followed by coupling with β -naphthol gives a dark blue coloured precipitate.

Reason :- It is an electrophilic aromatic substitution reaction.

(1) A (2) B (3) C (4) D

147. Assertion :- Benzaldehyde gives a positive test with Benedict's and Fehling solution.

Reason :- Benzaldehyde forms black precipitate with Fehling solution.

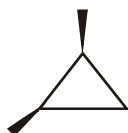
- (1) A (2) B (3) C (4) D

148. Assertion :- $\text{—}\overset{\text{O}}{\parallel}\text{C—Cl}$ is prior than $\text{—}\overset{\text{O}}{\parallel}\text{C—NH}_2$ in IUPAC naming.

Reason :- $\text{—}\overset{\text{O}}{\parallel}\text{C—Cl}$ is prior than $\text{—}\overset{\text{O}}{\parallel}\text{C—NH}_2$ according to CIP rules.

- (1) A (2) B (3) C (4) D

149. Assertion :- Given molecule is optically inactive.



Reason :- It has two chiral carbons.

- (1) A (2) B (3) C (4) D

150. Assertion :- Molecules those are not superimposable on their mirror image are chiral.

Reason :- All chiral molecules have one chiral carbon atom.

- (1) A (2) B (3) C (4) D

151. Assertion :- $\text{CH}_3\text{O—}\overset{\oplus}{\text{CH}}_2$ is more stable than $\text{CH}_3\text{—}\overset{\oplus}{\text{CH}}_2$.

Reason :- $\text{CH}_3\text{O—}$ has $-\text{I}$ effect.

- (1) A (2) B (3) C (4) D

152. Assertion :- Trifluoro acetic acid is more acidic than formic acid.

Reason :- $\text{CF}_3\text{—}$ group has strong $-\text{I}$ effect.

- (1) A (2) B (3) C (4) D

153. Assertion :- does not show tautomerism

Reason :- Enol of , will be highly unstable due to anti aromaticity.

- (1) A (2) B (3) C (4) D

154. Assertion :- $(\text{CH}_3)_3\text{C—CH=CH}_2$ does not show hyperconjugation.

Reason :- $\text{CH}_3\text{—CH=CH}_2$ shows hyper conjugation.

- (1) A (2) B (3) C (4) D

155. Assertion :- Acetone gives yellow precipitate of CHI_3 , when it is treated with $\text{I}_2 + \text{NaOH}$.

Reason :- CHI_3 is yellow in colour.

- (1) A (2) B (3) C (4) D

156. Assertion :- $\text{CH}_2=\text{CH—}\overset{\oplus}{\text{O}}\text{H}_2$ does not show resonance.

Reason :- O-atom can not form four bonds.

- (1) A (2) B (3) C (4) D

157. Assertion :- Isobutylene is an unsaturated compound.

Reason :- It has two double bond.

- (1) A (2) B (3) C (4) D

158. Assertion :- The IUPAC name for $\text{HC}\equiv\text{C—CH=CH}_2$ is but-3-en-1-yne.

Reason :- If there is a choice, a triple bond is always given a lower locant than a double bond.

- (1) A (2) B (3) C (4) D

159. **Assertion :-** $\begin{array}{c} \text{H} & & \text{Cl} \\ & \diagdown & / \\ & \text{C}=\text{C} \\ & / & \diagdown \\ \text{Cl} & & \text{Br} \end{array}$ and $\begin{array}{c} \text{H} & & \text{Br} \\ & \diagdown & / \\ & \text{C}=\text{C} \\ & / & \diagdown \\ \text{Cl} & & \text{Cl} \end{array}$
have Z and E configuration respectively.

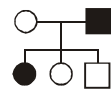
Reason :- They are cis and trans respectively.

- (1) A (2) B (3) C (4) D

160. **Assertion :-** $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H} - \text{C} - \text{CH}_3 \\ | \\ \text{H} \end{array}$ and $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H} - \text{C} - \text{H} \\ | \\ \text{CH}_3 \end{array}$
are conformer.

Reason :- First is less stable than second.

- (1) A (2) B (3) C (4) D

161. **Assertion :-**  This pedigree chart can not be for X linked dominant disorder.

Reason :- In X linked dominant disorder if mother is normal daughter can never be diseased.

- (1) A (2) B (3) C (4) D

162. **Assertion :-** Birds like rooster show male heterogamety

Reason :- Males produce [A + Z] and [A + W] gametes.

- (1) A (2) B (3) C (4) D

163. **Assertion :-** In f_2 generation of monohybrid cross, both parental traits are expressed in the proportion of 3 : 1.

Reason :- The contrasting parental traits show blending in f_2 generation.

- (1) A (2) B (3) C (4) D

164. **Assertion :-** The law of independent assortment can be studied by means of dihybrid cross.

Reason :- The law of independent assortment is applicable only for linked genes.

- (1) A (2) B (3) C (4) D

165. **Assertion :-** Male honey bee does not have father but have grand father.

Reason :- Male honey bee are produced by parthenogenesis is where as female honey bee are produced by fertilisation.

- (1) A (2) B (3) C (4) D

166. **Assertion :** UGG is a chain termination codon.

Reason : It does not code any amino acid.

- (1) A (2) B (3) C (4) D

167. **Assertion :** Wobbling process established the economy of t-RNA.

Reason : In wobbling process single t-RNA recognizes more than one codon of m-RNA.

- (1) A (2) B (3) C (4) D

168. **Assertion :-** If sequence of one strand of DNA is known then sequence in other strand can be predicted.

Reason :- Complementary base pairing is found in DNA.

- (1) A (2) B (3) C (4) D

169. **Assertion :-** Taylor and colleagues in 1958 used CsCl density gradient centrifugation technique to prove semiconservative DNA replication.

Reason :- They use ^{15}N heavy nitrogen which is not a radioactive isotope.

- (1) A (2) B (3) C (4) D

170. **Assertion :-** On template $5' \rightarrow 3'$ DNA replicate in discontinuous manner.

Reason :- DNA polymerase catalyse polymerisation only in one direction that is $5' \rightarrow 3'$.

- (1) A (2) B (3) C (4) D

171. Assertion :- Homologous organ show convergent evolution.

Reason :- Homologous organs show different ancestry.

- (1) A (2) B (3) C (4) D

172. Assertion :- First life originated in water.

Reason :- Condition are suitable in water for origin of life.

- (1) A (2) B (3) C (4) D

173. Assertion :- Mutations help in evolution.

Reason :- Mutation cause variations.

- (1) A (2) B (3) C (4) D

174. Assertion :- Potato and sweet potato are example of homologous organs.

Reason :- Homologous organs are result of divergent evolution.

- (1) A (2) B (3) C (4) D

175. Assertion :- The evolution of modern giraffe due to repeated use of their neck and forelimbs.

Reason :- Repeated use of neck is helpful in increase their reproductive capacity.

- (1) A (2) B (3) C (4) D

176. Assertion :- Natural selection is a mechanism for evolution.

Reason :- Mutation always expressed in phenotype.

- (1) A (2) B (3) C (4) D

177. Assertion :- Heart of vertebrates are example of homologous organ.

Reason :- Heart of vertebrates show convergent evolution.

- (1) A (2) B (3) C (4) D

178. Assertion :- Evolution is always directed process.

Reason :- we can always determine the direction of evolution.

- (1) A (2) B (3) C (4) D

179. Assertion :- Single step large mutation is called saltation.

Reason :- Natural mutation are random process.

- (1) A (2) B (3) C (4) D

180. Assertion :- Mutation can changed the Hardy-weinberg equilibrium.

Reason :- Mutation can change the gene frequency in a population.

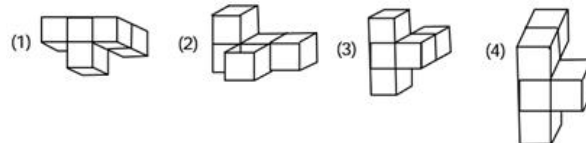
- (1) A (2) B (3) C (4) D

181. The solid so formed by joining unit cubes is rotated to obtain different positions, which of these cannot be the shape after it has turned ?

Question Figure :



Answer Figures :



- (1) 1 (2) 2 (3) 3 (4) 4

182. Consider the following.

Statements and Conclusions :

Statements :

Some rats are cats.

Some cats are dogs.

No dog is a cow

Conclusions :

No cow is a cat.

No dog is a rat.

Some cats are rats.

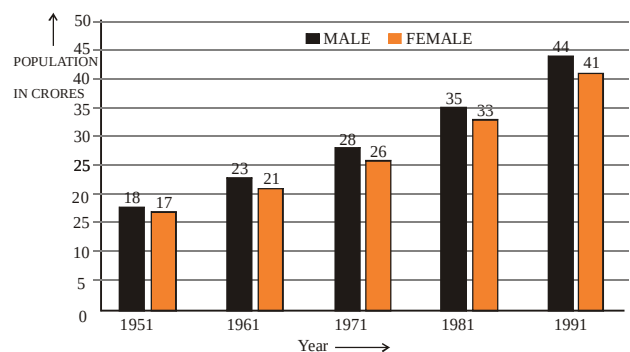
Which of the above conclusions is/are drawn from the statements ?

- (1) I, II and III (2) Only I and II
(3) Only III (4) Only II and III

- 183.** The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of four parallel lines, is
(1) 18 (2) 24 (3) 32 (4) 36
- 184.** In a school every student is assigned a unique identification number. A student is a football player if and only if the identification number is divisible by 4, whereas a student is a cricketer if and only if the identification number is divisible by 6. If every number from 1 to 100 is assigned to a student, then how many of them play cricket as well as football ?
(1) 4 (2) 8 (3) 10 (4) 12
- 185.** When a runner was crossing the 12 km mark, she was informed that she had completed only 80% of the race. How many kilometres was the runner supposed to run in this event ?
(1) 14 (2) 15 (3) 16 (4) 16.5
- 186.** Raju has 9000 with him and he wants to buy a mobile handset; but he finds that he has only 75% of the amount required to buy the handset. Therefore, he borrows 2000 from a friend. Then
(1) Raju still does not have enough amount to buy the handset.
(2) Raju has exactly the same amount as required to buy the handset.
(3) Raju has enough amount to buy the handset and he will have 500 with him after buying the handset.
(4) Raju has enough amount to buy the handset and he will have 1000 with him after buying the handset.

- 187.** In 2002, Meenu's age was one-third of the age of Meera, whereas in 2010, Meenu's age was half the age of Meera. What is Meenu's year of birth ?
(1) 1992 (2) 1994
(3) 1996 (4) 1998
- 188.** Rakesh and Rajesh together bought 10 balls and 10 rackets. Rakesh spent 1300 and Rajesh spent 1500. If each racket costs three times a ball does, then what is the price of a racket ?
(1) 70 (2) 90 (3) 210 (4) 240
- 189.** In a conference, out of a total 100 participants, 70 are Indians. If 60 of the total participants are vegetarian, then which of the following statements is/are correct ?
[a] At least 30 Indian participants are vegetarian.
[b] At least 10 Indian participants are non-vegetarian.
Select the correct answer using the codes given below :
(1) a only (2) b only
(3) Both a and b (4) Neither a nor b

The bar graph shows the number of males and females (in crores) in India during 1951-1991. Read the graph and answer the following questions



- 190.** What was the approximate percentage of female population in India during 1991 ?
(1) 48.23 (2) 48.02 (3) 48.03 (4) 48.33

191. What was the approximate number of males in India in 1971 per thousand females :-

- (1) 913 (2) 1075 (3) 1077 (4) 1175

192. A certain number of men can do a piece of work in 60 days. If there were 6 men more, the work can be finished 20 days earlier. The number of men working is :-

- (1) 6 (2) 12 (3) 18 (4) 24

193. For the following questions

Find the odd word/letters/number from the given alternatives.

- (1) BDGK (2) JLOS (3) NPSW (4) MORU

194.  :  ::  : ? :-

Statue of Unity

Spring temple

(1)



(2)



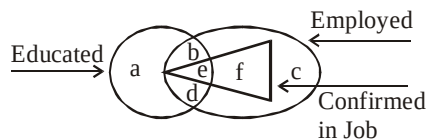
(3)



(4)



195. Study the following Venn Diagram and find the region representing persons who are educated and employed but not confirmed.



- (1) a, c (2) a, b, c (3) b, d (4) a, d, c

196. Choose the correct water image of the question figure out of four answer figures given:

Question Figure :



Answer Figures :



(1) 1

(2) 2

(3) 3

(4) 4

197. Who among the following launched the 'Fit India Movement' ?

- (1) Amit Shah (2) Narendra Modi
(3) Smriti Irani (4) Rajnath Singh

198. India is buying 'Spice-2000' bombs from which of the following country ?

- (1) Russia (2) France
(3) USA (4) Israel

199. Feroz Shah Kotla Stadium in Delhi will be renamed as _____.

- (1) Sushma Swaraj Stadium
(2) Ram Manohar Lohia Stadium
(3) Vallabhbhai Patel Stadium
(4) Arun Jaitley Stadium

200. Relate the following with respect to 12th Cricket World Cup Semifinals :-

